



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to PKS FUELS UNIT II (COCO POCHAMPALLI)

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

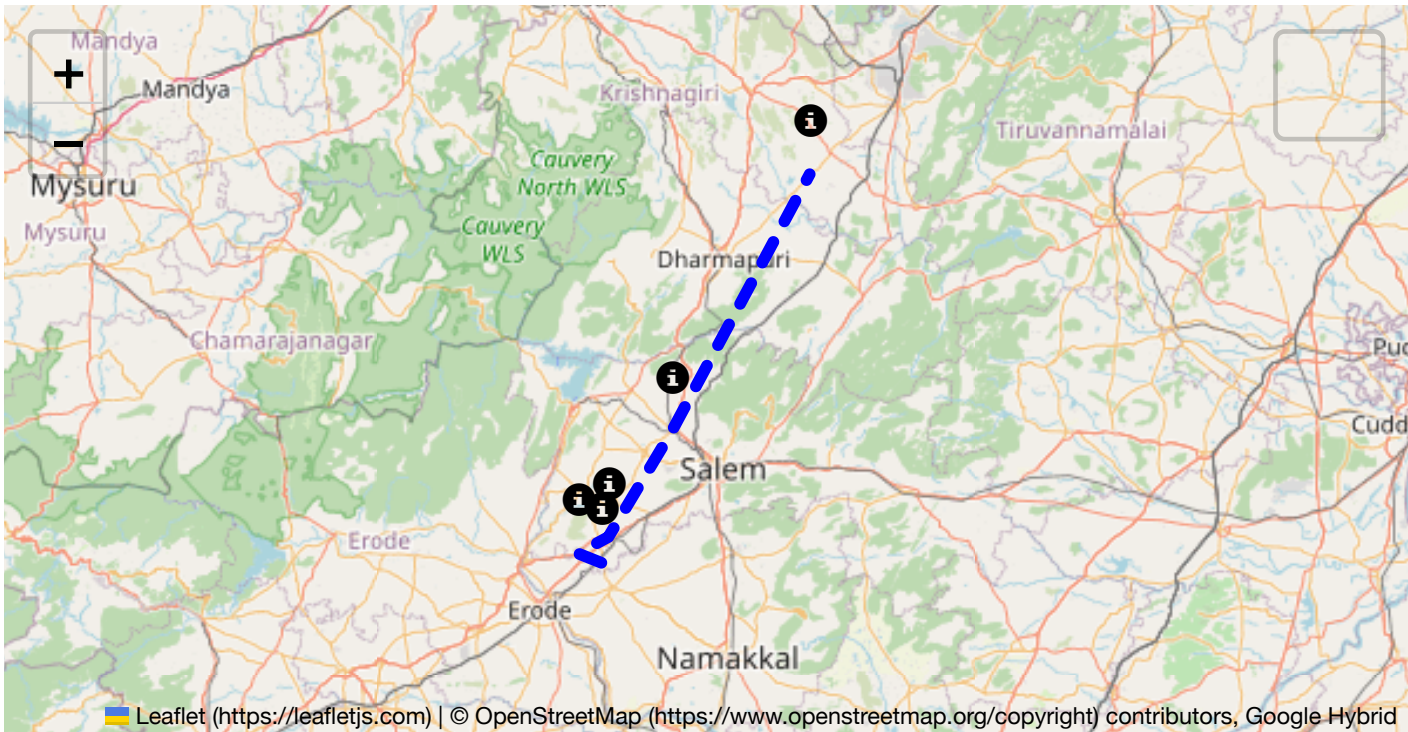
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

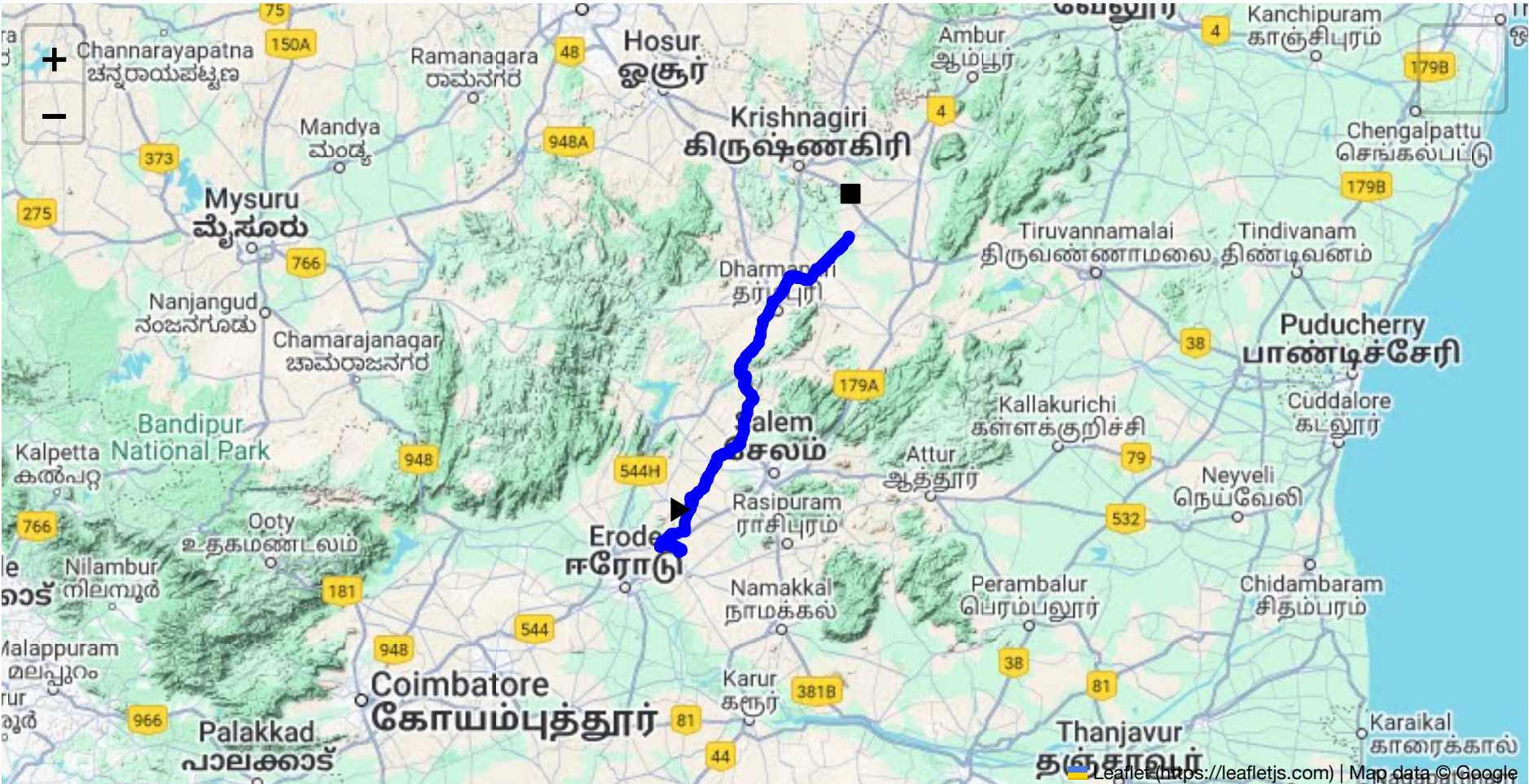


Route Summary:
Total Distance: 141.37 km
Estimated Duration: 3.0 hours

Adjusted Duration (Heavy Vehicle): 3.7 hours

Start: (11.4381, 77.8734)

End: (12.329381, 78.362189)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Pochampalli, passing through Bhavani Main Road, SH 86, and Salem, covers about 141.37 kilometers. This journey travels through rural and semi-urban parts of Tamil Nadu, using primarily state highways and some local roads. The route is moderately direct, connecting major local points and communities. It passes through several towns and relies heavily on well-used arterial roads.

2. Weather Conditions and Potential Hazards

Tamil Nadu generally experiences a tropical climate, with significant heat and occasional heavy rains during the monsoon seasons (June to September and October to November). The route may be susceptible to heavy rain, leading to slippery roads and possible flooding in low-lying areas, especially during peak monsoons. Drivers should be cautious about fog during early mornings in winter months, affecting visibility.

3. Traffic Patterns

- **Sangagiri to Salem:** Moderate traffic, with peak hours typically in the morning (7:30-9:00 AM) and evening (5:00-7:00 PM).
- **Salem:** As a major city, Salem experiences heavy congestion, especially during peak hours (8:00-10:00 AM and 5:30-8:00 PM). Key congestion zones are near marketplaces and commercial areas.
- **Salem to Pochampalli:** This segment may experience lighter traffic, though irregular congestion can occur due to local events or roadworks.

4. Road Quality and Infrastructure

The route consists mainly of state highways that are generally in fair condition. However, certain segments, especially rural stretches, may experience potholes or narrower lanes that could affect heavy vehicles. Ongoing improvements or works might also present temporary challenges.

5. Alternative Routes for Emergencies

In case of roadblocks or emergencies, alternative routes through lesser-known local roads connecting towns like Tharamangalam or Omalur might be utilized. However, these should be vetted for road conditions and accessibility, especially for heavy vehicles.

6. Local Regulations on Hazardous Material Transport

Handling and transport of hazardous materials require stringent adherence to national safety guidelines, including proper labeling and documentation. Multiple check-posts, particularly near Salem, necessitate compliance with safety regulations.

7. Historical Incidents

Past incidents have often involved vehicular collisions around Salem due to congestion and road conditions. However, specific historical incidents involving hazardous materials are rare but require constant vigilance due to urban proximity near Salem.

8. Environmental Considerations and Sensitive Areas

The route passes through ecologically sensitive zones, particularly around bodies of water and agricultural land. Careful management of emissions and avoidance of spillages are crucial to minimize environmental impact.

9. Communication Coverage and Dead Zones

Mobile network coverage along this route is generally good in urban and semi-urban areas. However, remote segments may experience brief dead zones. Drivers should plan communication needs accordingly.

10. Estimated Emergency Response Times

- **Sangagiri to Salem:** Approx. 30-45 minutes
- **Within Salem:** Approx. 15-25 minutes, considering urban proximity to emergency services
- **Salem to Pochampalli:** 30-60 minutes, influenced by rural distances

11. Overall Summary of Risk Assessment

This route is moderately safe for transporting hazardous materials with adherence to regulations. Key risks include weather-related and congestion-induced delays and moderate road conditions. Regular checks and communication plans, with awareness of environmental considerations, ensure safe transit. Emergency preparedness is essential for immediate response to unforeseen situations. Continual monitoring of roadworks and compliance with local regulations will contribute to safer operations.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45781, 77.85203	30 KM/Hr
5	Turn	High	11.46307, 77.84941	15 KM/Hr
6	Turn	High	11.45542, 77.81725	15 KM/Hr
7	Turn	High	11.45605, 77.81683	15 KM/Hr
8	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
9	Turn	High	11.49469, 77.88488	15 KM/Hr
10	Turn	Medium	11.55618, 77.89789	30 KM/Hr
11	Turn	Medium	11.55699, 77.89986	30 KM/Hr
12	Turn	High	11.58228, 77.90841	15 KM/Hr
13	Turn	High	11.58340, 77.90637	15 KM/Hr
14	Turn	Medium	11.59201, 77.90649	30 KM/Hr
15	Turn	Medium	11.63882, 77.94859	30 KM/Hr
16	Turn	High	11.64226, 77.95200	15 KM/Hr
17	Turn	Medium	11.64716, 77.95118	30 KM/Hr
18	Turn	Medium	11.67868, 77.96836	30 KM/Hr
19	Turn	Medium	11.70077, 77.97957	30 KM/Hr
20	Turn	High	11.70223, 77.97860	15 KM/Hr
21	Turn	Medium	11.71402, 77.99586	30 KM/Hr
22	Turn	Medium	11.71704, 77.99948	30 KM/Hr
23	Turn	Medium	11.71918, 78.01127	30 KM/Hr
24	Turn	Medium	11.72708, 78.02197	30 KM/Hr
25	Turn	Medium	11.73769, 78.03451	30 KM/Hr
26	Turn	Medium	11.73785, 78.03551	30 KM/Hr
27	Turn	High	11.73651, 78.03871	15 KM/Hr
28	Turn	Medium	11.74271, 78.04717	30 KM/Hr
29	Turn	Medium	11.86847, 78.08367	30 KM/Hr
30	Turn	Medium	11.87942, 78.08197	30 KM/Hr
31	Turn	Medium	11.92585, 78.06507	30 KM/Hr
32	Turn	Medium	11.93704, 78.05402	30 KM/Hr
33	Turn	High	11.95818, 78.04978	15 KM/Hr
34	Turn	Medium	11.95988, 78.04948	30 KM/Hr
35	Turn	Medium	11.96304, 78.05136	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
36	Turn	High	11.96399, 78.05135	15 KM/Hr
37	Turn	High	11.96580, 78.04923	15 KM/Hr
38	Turn	Medium	11.96675, 78.04939	30 KM/Hr
39	Turn	Medium	11.96791, 78.05070	30 KM/Hr
40	Turn	Medium	12.19881, 78.18157	30 KM/Hr
41	Turn	High	12.21894, 78.19193	15 KM/Hr
42	Turn	Medium	12.21814, 78.19468	30 KM/Hr
43	Turn	High	12.21363, 78.21151	15 KM/Hr
44	Turn	High	12.21453, 78.21202	15 KM/Hr
45	Turn	Medium	12.21328, 78.21498	30 KM/Hr
46	Turn	Medium	12.21249, 78.21555	30 KM/Hr
47	Turn	Medium	12.21122, 78.22046	30 KM/Hr
48	Turn	Medium	12.21146, 78.22233	30 KM/Hr
49	Turn	High	12.21114, 78.22332	15 KM/Hr
50	Turn	High	12.21037, 78.22363	15 KM/Hr
51	Turn	High	12.20521, 78.24447	15 KM/Hr
52	Turn	Medium	12.26553, 78.30335	30 KM/Hr
53	Turn	Medium	12.26772, 78.30415	30 KM/Hr
54	Turn	Medium	12.26941, 78.30683	30 KM/Hr
55	Turn	Medium	12.28415, 78.32000	30 KM/Hr
56	Turn	Medium	12.28483, 78.32243	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
8	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
11	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
13	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
14	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
15	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
16	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
20	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
21	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
24	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
25	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
9	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
22	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55618, 77.89789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58340, 77.90637



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59201, 77.90649



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63882, 77.94859



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64226, 77.95200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64716, 77.95118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67868, 77.96836



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70077, 77.97957



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70223, 77.97860



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71402, 77.99586



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71704, 77.99948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71918, 78.01127



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72708, 78.02197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73769, 78.03451



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73785, 78.03551



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03871



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74271, 78.04717



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86847, 78.08367



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92585, 78.06507



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93704, 78.05402



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95988, 78.04948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19881, 78.18157



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21894, 78.19193



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.21814, 78.19468



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21363, 78.21151



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21453, 78.21202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.21328, 78.21498



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.21249, 78.21555



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.21146, 78.22233



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21114, 78.22332



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21037, 78.22363



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.20521, 78.24447



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.26553, 78.30335



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.26772, 78.30415



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.26941, 78.30683



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.28415, 78.32000



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.28483, 78.32243

